
Applied Cyber Threat Analysis – NMAP/Nessus Project

- This lab was conducted within the Virginia Cyber Range Environment.
 - Report based off assignment assigned by George Mason University - Department of Information Sciences and Technology College of Engineering and Computing
 - Cheche Agada. (2024). Mini Project IT-462: Applied Cyber Threat Analysis, George Mason University
-

Introduction:

This lab will guide you through the process of installing a hypervisor (VMWare), and two virtual machines – Windows and bee-box.

bee-box is a **custom Linux VM pre-installed with bWAPP (buggy web application)**. It is intentionally pre-installed with a vulnerable application to provide the opportunity to explore application vulnerabilities in a test environment. The VM gives several opportunities to hack and deface the bWAPP website that is contained within it. However, this exercise will only focus on assessing the vulnerabilities, not exploiting them.

You will use Nmap/ZenMap and Nessus to scan applications and systems for specific vulnerabilities like open ports and other software vulnerabilities.

Objective:

Set up a virtual environment on your personal device and conduct vulnerability assessment exercises.

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Part 1 - Setup

Task 1 – Install VMWare

Task 2 – Install a Windows Virtual Machine

Lab Exercise 1

- VMWare version **17.6.3**

VMware Workstation Pro (For Windows)					Release 17.6.3
File Name	Release Date	Last Updated	SHA2	MD5	
VMware Workstation Pro for Windows					
VMware-workstation- full-17.6.3-24563834.exe(401.43 MB) Build Number: 24563834	Mar 04, 2025	Feb 24, 2025	d7c04b4dd1e6bf551693897d4805e90c45198a830c5361 d8af6267b40906857b	de562b18a39513e3414ff197ec1a4cb1c	Download

Windows 10 Education, version 22H2

Microsoft Azure

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windows

Product category :

Product language :

29 Items

Name ↑↓	Product category ↑↓	Operat
Machine Learning Serv...	AI + Machine Learning	Windon
Machine Learning Serv...	AI + Machine Learning	Windon
Machine Learning Serv...	AI + Machine Learning	Windon
Microsoft R Server 9.1...	Database	Windon
Windows 10 Education...	Operating System	Windon
Windows 10 Education...	Operating System	Windon
Windows 11 Education...	Operating System	Windon
Windows 11 Education...	Operating System	Windon
Windows 10, version 2...	Operating System	Windon

Software

Education

Windows 10 Education, version 22H2

Windows 10 Education builds on Windows 10 Enterprise and provides enterprise-grade manageability and security for schools. Windows 10 Education is effectively a variant of Windows 10 Enterprise that provides education-specific default settings.

Operating System

Windows

Product language

English

System

64 bit

Product key

VMRPM-C9NG4-J66MQ-2GYRD-T6PJB

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PART 2

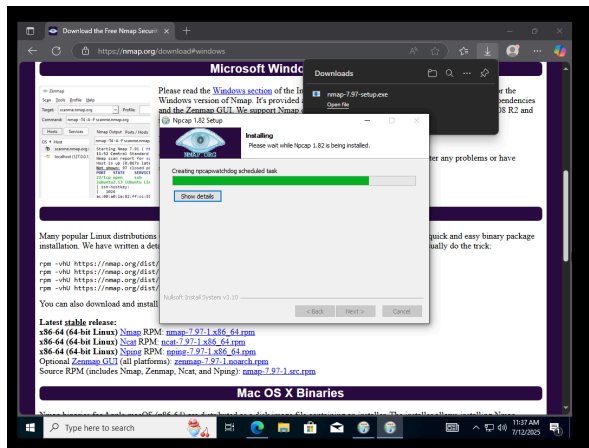
Overview

In this section, you will conduct security assessment tasks to check for system and software vulnerabilities. We will use Nmap/ZenMap to check for open ports and services in order to determine vulnerable or exploitable services. Then we will use Nessus to scan a buggy application installed on a virtual machine to check for vulnerabilities in the application.

Remember to include screenshots to show each step.

Task 3A: Conduct an Nmap Scan of Your Host Computer using ZenMap

- Download and install Nmap on your Windows virtual machine (VM) (<https://nmap.org/download>)
- ZenMap is the GUI version of Nmap – they are installed together.



- On your host computer, open a command prompt and obtain the IP (Internet Protocol) address.

```
Command Prompt
Microsoft Windows [Version 10.0.26100.4652]
(c) Microsoft Corporation. All rights reserved.

C:\Users\troy>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

   Media State . . . . . : Media disconnected
   Connection-specific DNS Suffix  . : 

Ethernet adapter Ethernet 4:

   Connection-specific DNS Suffix  . : 
   Link-local IPv6 Address . . . . . : fe80::5c62:bc66:3adc:5c39%6
   IPv4 Address. . . . . : 192.168.56.1
   Subnet Mask . . . . . : 255.255.255.0
   Default Gateway . . . . . : 

Wireless LAN adapter Local Area Connection* 1:

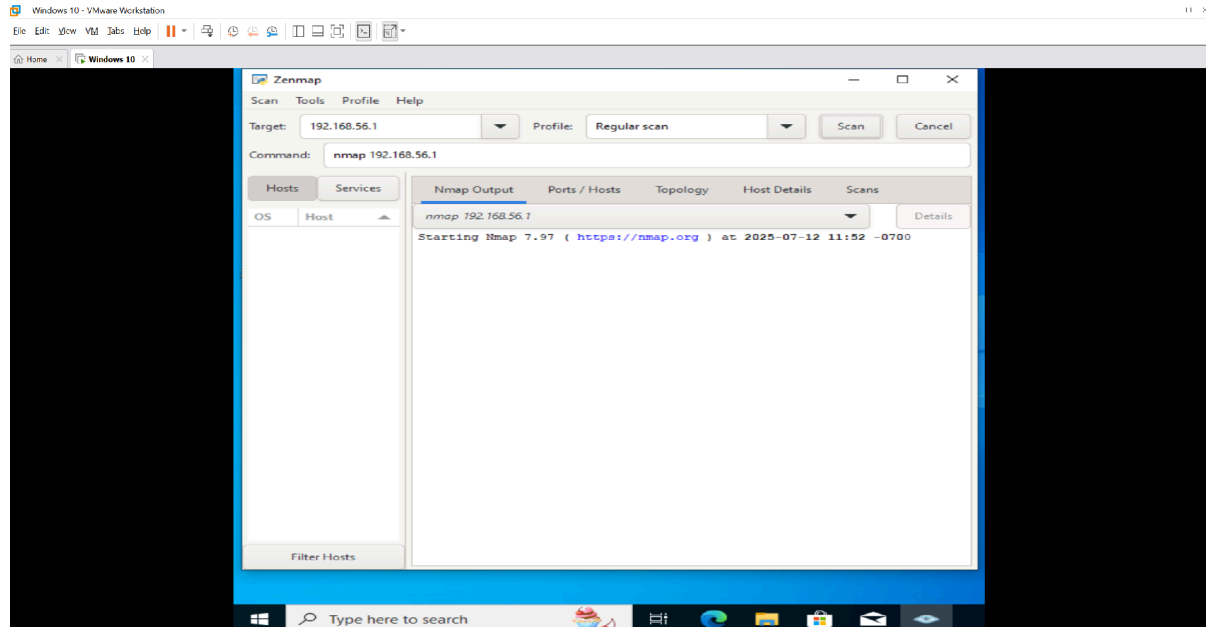
   Media State . . . . . : Media disconnected
   Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . : Media disconnected
```

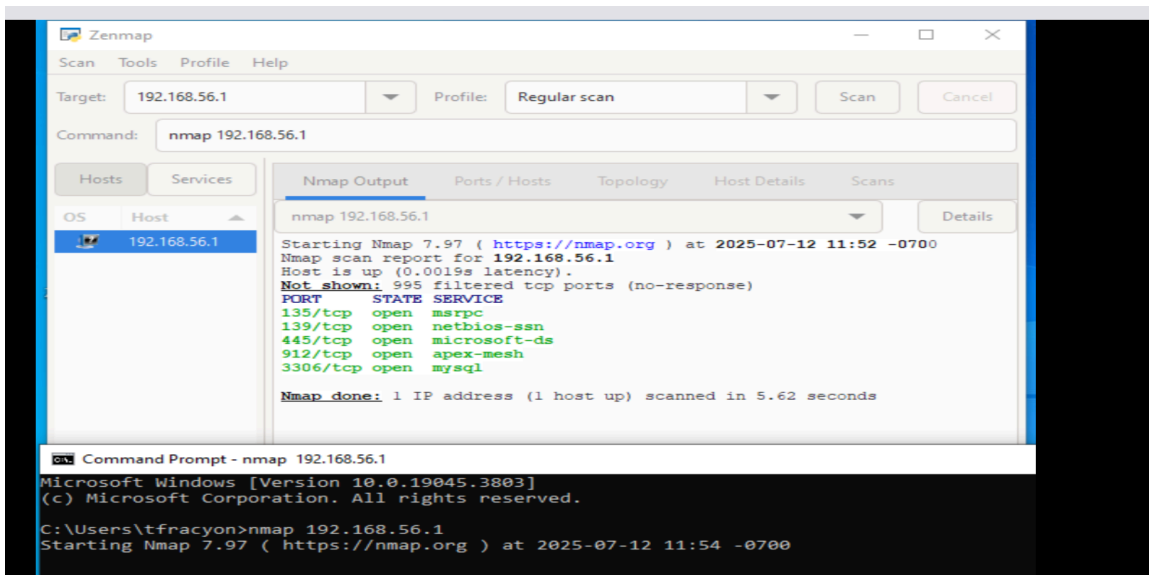
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- On your VM, start the Nmap program.
- Provide the IP address of your host machine in the space provided (**Target**).
- In the **Profile** section, select “Regular scan” as the type of scan to conduct.
- Nmap generates the required command as shown in the **Command** section.
- Select the **Scan** button to start the scan.



Task 3B: Conduct an Nmap Scan of Your Host Computer (Command Line)

- Run the same command from (3A) on the command prompt of your VM using Nmap.



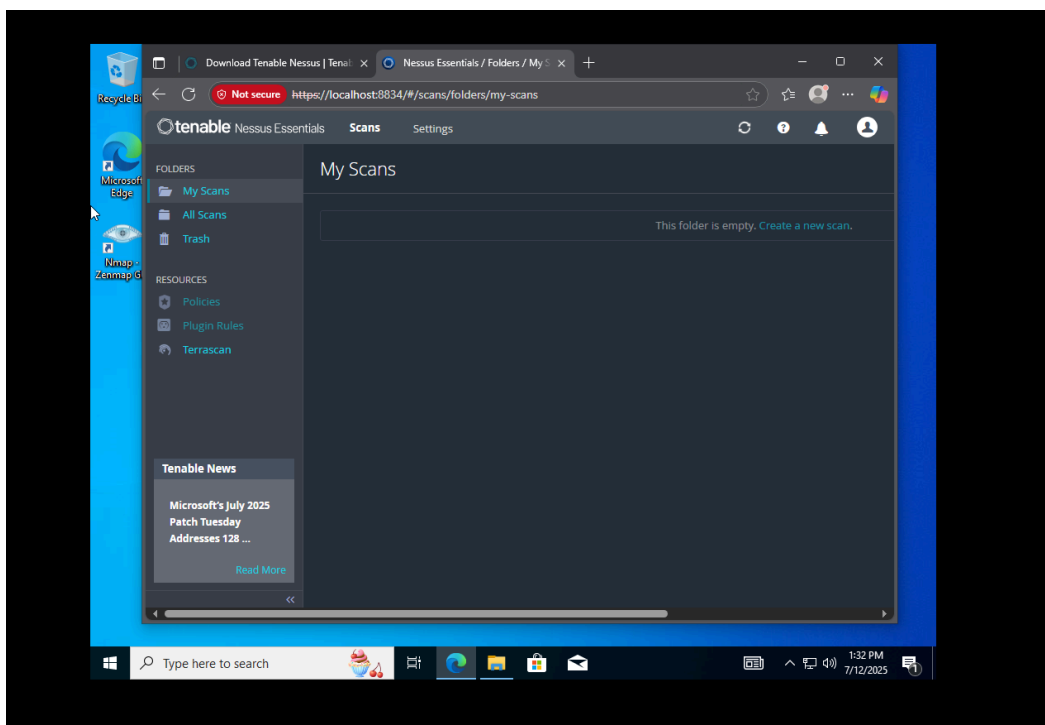
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Lab Exercise 2

- (1) Compare the output of 3A and 3B. Are they the same? Why or why not?
They are the same because they both list the port #, the state of the service, and the name of the service.
- (2) How many ports are open? **5 ports**
- (3) Select an open port and investigate the service hosted at this port. Is it safe to have the selected port open? What risk is associated with having this port open?
Port 3306 is associated with mysql. It is considered unsafe to have this port left open because of unauthorized personnel getting access.
- (4) How many ports are filtered? **995 filtered tcp ports.**
- (5) What does a filtered port mean? **A firewall is blocking access.**

Task 4: Install Nessus

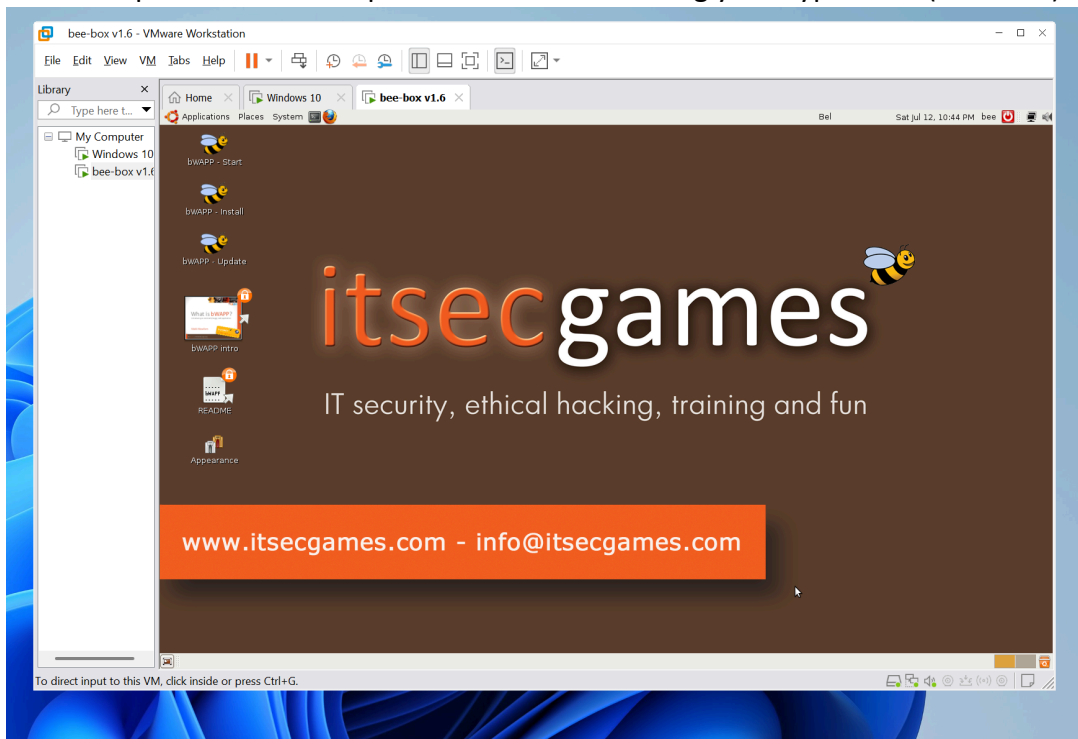
- Download and install Nessus Essentials on your Windows VM (<https://www.tenable.com/products/nessus/nessus-essentials>)
- Program will run in your browser (<https://localhost:8834>)



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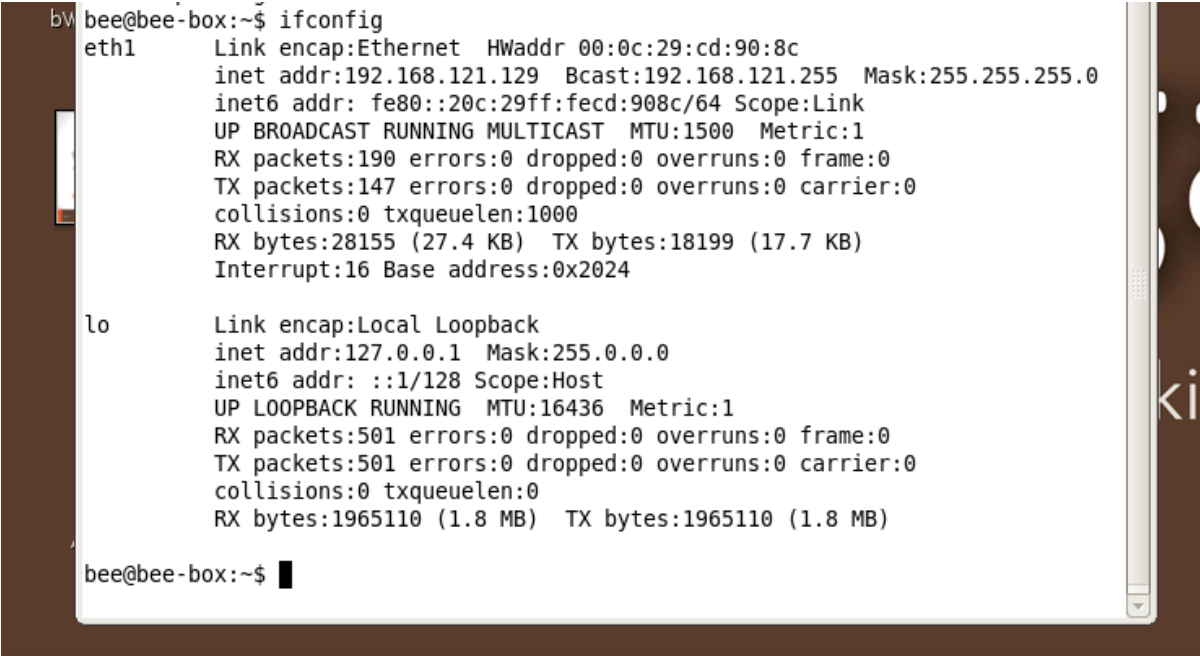
Task 5: Set up bee-box

- Download the buggy Web Application (bWAPP) VM – **bee-box** – **directly to your computer**. You must download it from the **Mini Project** folder on Blackboard.
- Unzip the folder and open the bee-box VM using your hypervisor (VMWare).



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- If requested, provide username (**bee**) and password (**bug**) to login.
- Open a terminal window and change the language preference (i.e. **System** → **Preferences** → **Keyboard** → **Layouts**) to USA. You may have to remove and re-add 'USA' for your keyboard to work correctly.
- Obtain the IP address of the bee-box using '**ifconfig**' on the terminal.



```
bee@bee-box:~$ ifconfig
eth1      Link encap:Ethernet  HWaddr 00:0c:29:cd:90:8c
          inet addr:192.168.121.129  Bcast:192.168.121.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fe8c:908c/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:190 errors:0 dropped:0 overruns:0 frame:0
          TX packets:147 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:28155 (27.4 KB)  TX bytes:18199 (17.7 KB)
          Interrupt:16 Base address:0x2024

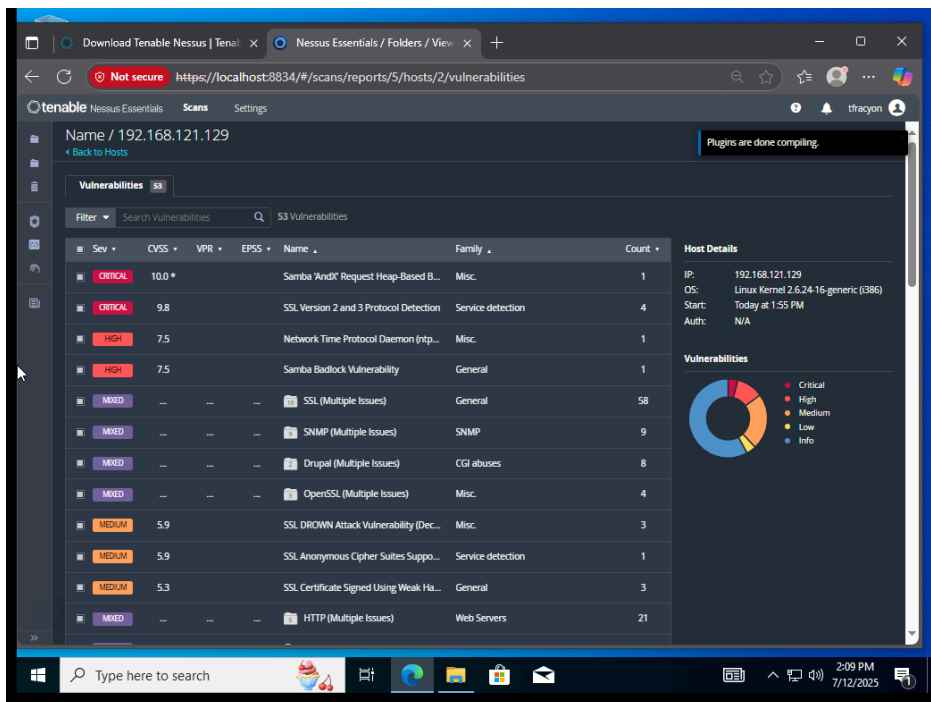
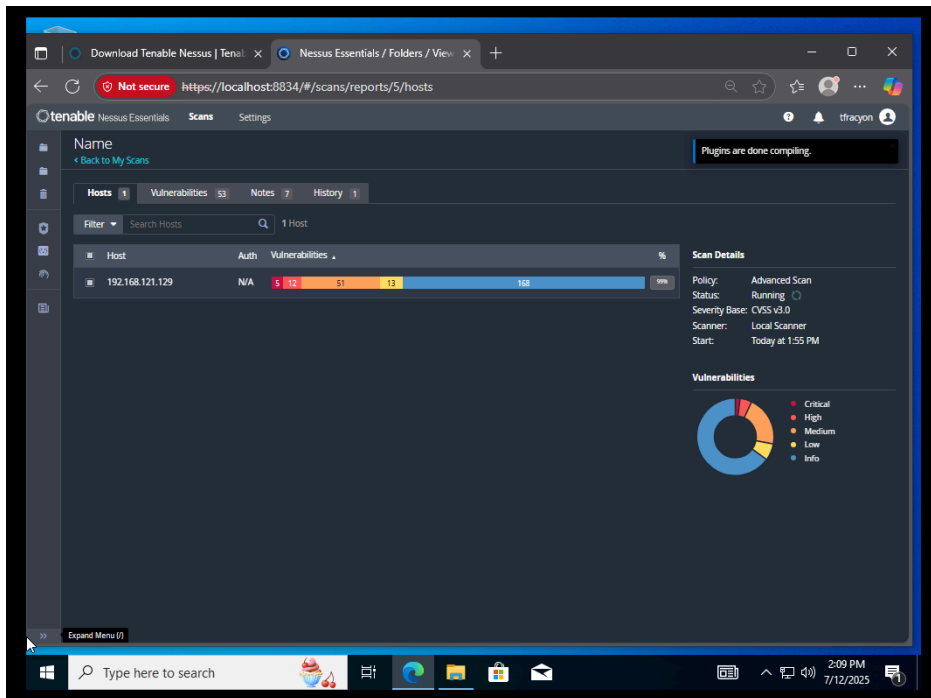
lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:501 errors:0 dropped:0 overruns:0 frame:0
          TX packets:501 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:1965110 (1.8 MB)  TX bytes:1965110 (1.8 MB)

bee@bee-box:~$
```

Task 6A: Conduct an Unauthenticated Nessus Scan

- Start the Nessus scanner in your browser (<https://localhost:8834>)
- Select New Scan -> Advanced Scan
- Provide a name for your scan "**Name**"
- Provide the IP address of the VM to scan (i.e. IP address of the bWAPP VM) "**Targets**"
- Click the arrow on the right side of the "**Save**" button and select "**Launch**" to start the scan
- After the scan is complete, examine the report.

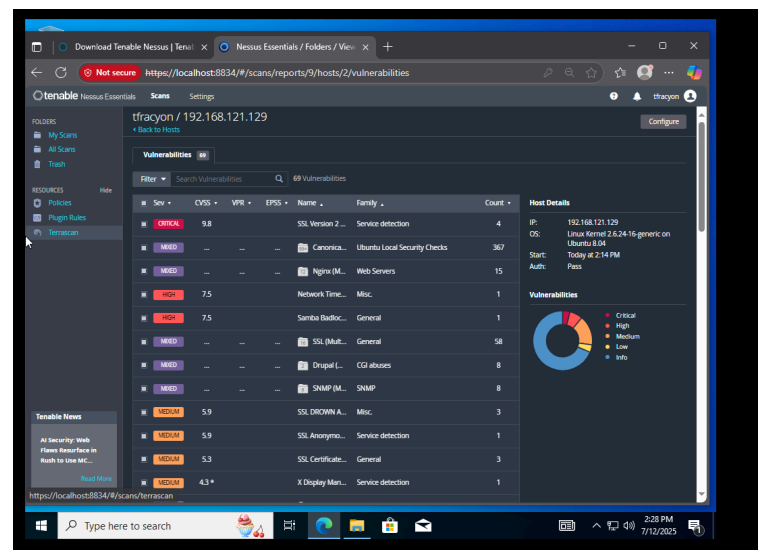
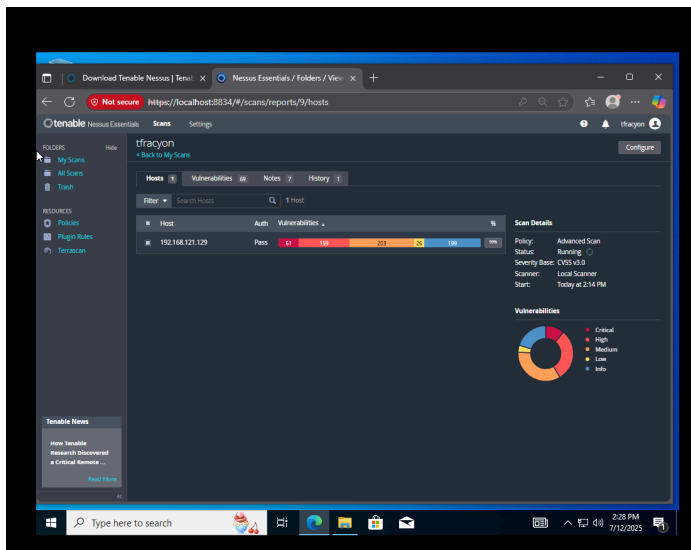
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Task 6B: Conduct an Authenticated Nessus Scan

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- Start the Nessus scanner in your browser (localhost:8834)
- Select **New Scan -> Advanced Scan**
- Provide a name for your scan **"Name"**
- Provide the IP address of the VM to scan (i.e. IP address of the bWAPP VM) **"Targets"**
- Click on the **"Credentials"** tab and select **"SSH"**
- Select **"password"** from the drop-down menu beside **"Authentication method"**
- Provide the username (**bee**) and password (**bug**)
- Select **"sudo"** from the drop down beside **"Elevate privileges with"**
- Click the arrow on the right side of the **"Save"** button and select **"Launch"** to start the scan
- After the scan is complete, examine the report.



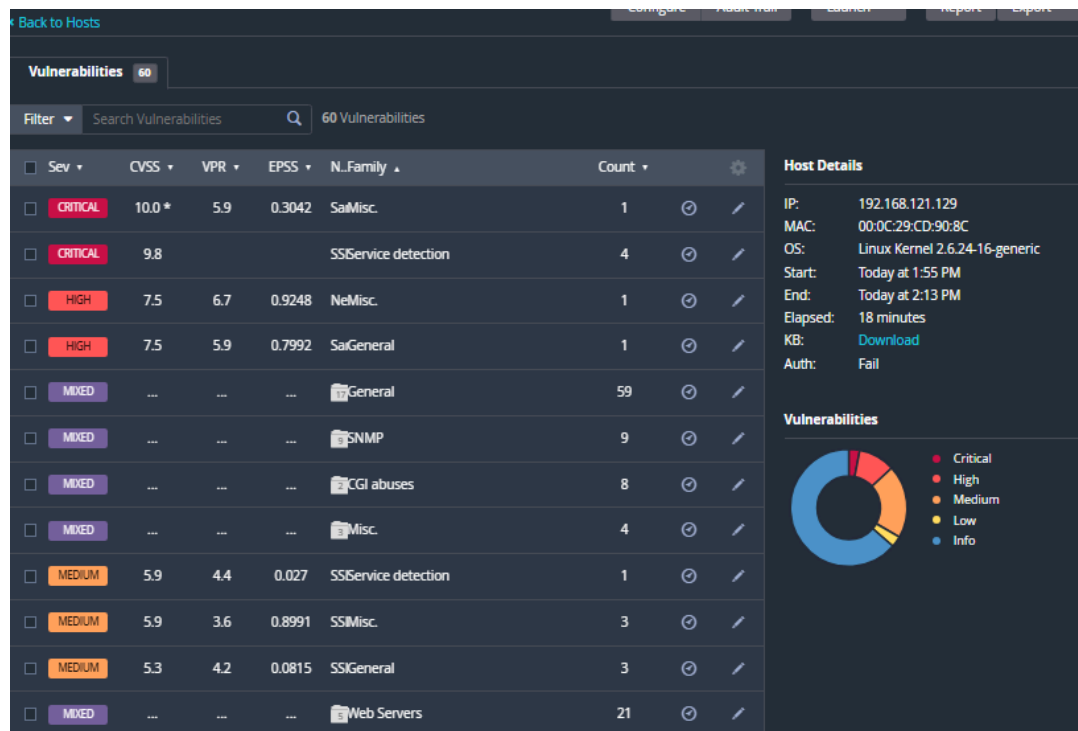
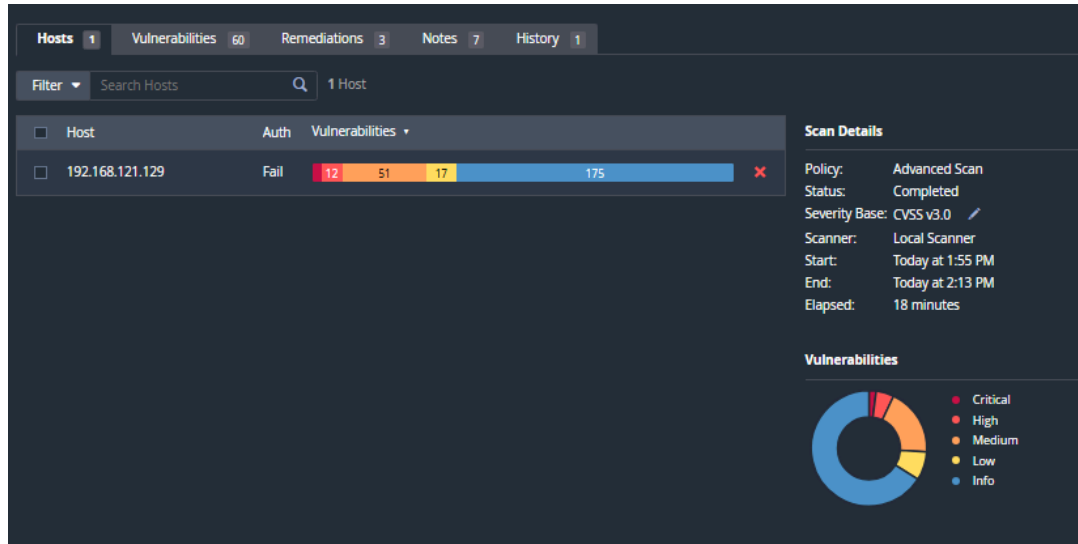
Lab Exercise 3

- (1) Compare the output of Task 6A and Task 6B. Are they the same? Why or why not?

They are not the same, there is a significant increase in the numbers of all types of vulnerabilities. There is a larger ratio of the more severe vulnerabilities. There are also more remediation recommendations on the 6B scan.

- (2) Include screenshots for the outputs for Task 6A and Task 6B

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Configure Audit Trail Launch Report Export

Back to My Scans

Hosts 1 Vulnerabilities 81 Remediations 125 Notes 7 History 1

Filter Search Hosts 1 Host

Host	Auth	Vulnerabilities
192.168.121.129	Pass	63 165 206 30 205

Scan Details

Policy: Advanced Scan
Status: Completed
Severity Base: CVSS v3.0
Scanner: Local Scanner
Start: Today at 2:14 PM
End: Today at 2:32 PM
Elapsed: 19 minutes

Vulnerabilities

Legend: Critical, High, Medium, Low, Info

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Back to My Scans

Hosts 1 Vulnerabilities 81 Remediations 125 Notes 7 History 1

Filter Search Vulnerabilities 81 Vulnerabilities

Sev	CVSS	VPR	EPSS	N.Family	Count		
CRITICAL	10.0 *	5.9	0.3042	SaMisc.	1		
CRITICAL	9.8	9.5	0.9422	BaGain a shell remotely	1		
CRITICAL	9.8			SSService detection	4		
MIXED	...	...	...	Ubuntu Local Security Checks	367		
MIXED	...	...	...	Web Servers	15		
HIGH	7.5	6.7	0.9248	NeMisc.	1		
HIGH	7.5	5.9	0.7992	SaGeneral	1		
MIXED	...	...	...	General	59		
MIXED	...	...	...	SNMP	9		
MIXED	...	...	...	CGI abuses	8		
MIXED	...	...	...	Misc.	5		
MEDIUM	5.9	4.4	0.027	SSService detection	1		

Scan Details

Policy: Advanced Scan
Status: Completed
Severity Base: CVSS v3.0
Scanner: Local Scanner
Start: Today at 2:14 PM
End: Today at 2:32 PM
Elapsed: 19 minutes

Vulnerabilities

Legend: Critical, High, Medium, Low, Info

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Resources

- Creating Windows 10 VM in VMWare
<https://www.youtube.com/watch?v=Gf2-p7VzbMg>
https://www.youtube.com/watch?v=O4-9YJD_XNg
- Nessus
<https://www.youtube.com/watch?v=rSGGrbpL7dc>
<https://www.youtube.com/watch?v=xJ9k1ksgA9Y>
https://www.youtube.com/watch?v=ElqyY_3f0zk